



▪ Constituent College of JSS Science and Technology University

▪ Approved by A.I.C.T.E

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

24CS111/24CS211 - C PROGRAMMING

TEST-1 QUESTION BANK ON UNIT-1 AND 2 (TOPIC-WISE)

Unit-1: Introduction to Computing, Overview of C, Constants, Variables & Data Types	
1.	Know the structure of a computer and have an idea about its functions/Explain the main components/functional units of a Computer System with a block diagram. (<i>Components of a Computer</i>)
2.	What are computer hardware and software? Briefly explain with examples. /Discuss the Hardware & Software with examples. (<i>Hardware and Software of a Computer</i>)
3.	Learn to develop programming logic through Algorithms and Flowcharts. / Define algorithms and flowcharts. / How are algorithms and flowcharts used to develop programming logic? Give examples. / What is an algorithm? Explain its importance and provide a simple example. Describe the purpose of a flowchart. / Discuss the notations used in describing algorithms and symbols used in flowcharts. (<i>Algorithm and Flowchart</i>)
4.	Discuss the genesis / history of C programming language. / Trace the history of the C programming language. How has C contributed to modern programming? / Produce an overview/importance of C programming language. (<i>History and Importance of C</i>)
5.	Describe the basic structure of C program with example. / As per ANSI C. how many sections are identified? Explain with an example program. / Write a C program to demonstrate any four C sections as per ANSI C standards. (<i>Basic structure of C program,)</i>
6.	Describe how C programs are created, compiled and executed with an example. / Describe the process of creating and executing a C program under UNIX system with example. (<i>Executing a C program</i>)
7.	Describe the various data types and their ranges. / Describe the basic data types. How could we extend the range of values they represent? / Differentiate between the data types int, float, and double in C with example. / Describe the purpose of following keywords: <i>short, long, signed & unsigned</i> (<i>Data Types</i>)
8.	Describe all the C tokens with examples for each. / What are C tokens? Explain with examples. (<i>C Tokens</i>)
9.	Discuss how variables are declared and their values are assigned. / How do you declare and initialize a variable in C? / Discuss how variables are used in a program. (<i>Declaration of Variables</i>)
10.	Illustrate the use of symbolic constants through sample program. / State the purpose of variable declaration and symbolic constant definition. / Explain how to define and use symbolic constants in C. (<i>Defining Symbolic Constants</i>)



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Unit-2: Operators and Expressions, Managing Input and Output Operations

1.	Know the various built-in operators of C language. (<i>Operators</i>) a. What are operators? With examples, explain the various types of operators in C. / Define operators and explain the different types of operators in programming with examples. / Identify the bitwise and Special operators. / What are relational operators? Write a program to compare two numbers and display whether they are equal, greater, or smaller. / What is a conditional operator? Write a program using conditional operators to find the largest of three numbers. b. Explain bitwise operators in detail with examples. Write a program to demonstrate bitwise AND, OR, and XOR operations. / Explain assignment operators and short-hand/compound assignment operators with examples. Write a program to demonstrate their usage. / What are increment and decrement operators? Explain their types (pre-increment and post-increment) with examples. / Explain special operators in detail with examples. Write a C program to demonstrate the same. / Describe logical operators. Write a program to demonstrate the use of &&, , and ! operators in decision-making.
2.	Explain the types of arithmetic operators. / Explain arithmetic operators with examples. Write a program to calculate the sum, difference, product, and quotient of two numbers. Evaluate the following expressions: (<i>Arithmetic Expressions</i>) i. $x = 9 - 12 / 3 + 3 * 2 - 1$ ii. $y = 9 - 12 / (3 + 3) * (2 - 1)$ iii. $z = 9 - (12 / 3 + 3) * 2 - 1$ iv. $l = 12 + 3 - 4 / 2 < 3 + 1$ v. $m = 9 - (12 / (3 + 3) * 2) - 1$ vi. $n = 9 - ((12/3) + 3 * 2) - 1$
3.	Explain type conversions in expressions. / What are implicit and explicit conversions? Give examples. / With suitable examples, explain the purpose of type conversions (both implicit and explicit) in C. / Write a c program to illustrate both type conversions. Evaluate w.r.t type conversion: int i, x; float f; double d; long int li; $x = li / i + i * f - d$; (<i>Type Conversions in Expressions</i>)
4.	Discuss how operator precedence and associativity rules are applied. / State the rule of precedence and associativity of arithmetic operators. / What is operator precedence and associativity? Explain with examples and a table of precedence for common operators. / With an example, explain how arithmetic expressions are evaluated. / State the rules for evaluation of expressions with suitable example. (<i>Operator Precedence and Associativity</i>)
5.	Describe how a character is read and written. / Illustrate any five-character test functions. (<i>Reading/Writing a Character</i>)
6.	Explain how formatted inputs are read with examples. / What are formatted input and output? Demonstrate the use of formatted input (scanf) and output (printf) in a program. / How can the readability of output be enhanced? Give examples. / Illustrate any five commonly used scanf() and printf format codes/specifiers/type specifiers/control strings. / Write a program to demonstrate the use of format specifiers (e.g., %d, %f, %c, %s) in input and output operations. (<i>Formatted Input/Output</i>)
7.	Differentiate putchar() and printf() functions. / Distinguish between getchar() and scanf() functions. / Mention the purpose of gets() and puts() functions.